Number: P3378R1
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constexpr exception types

This paper is mechanical wording change; making all exception types associated with constexpr compatible functionality marked `constexpr`. This is needed for consistency across library since allowing exception handling during constant evaluation by P3068.

As every exception type is just an ordinary type, they should be make constexpr compatible.

Changes

- R0 → R1: Updated status of `<exception>`'s exceptions, added note for library test macro (also in ...)

TL;DR

Now when we can throw exceptions during constant evaluation we should make sure all constexpr compatible functionality (eg. `vector`) is able to throw its exception types (eg. `out_of_range`) and allow users to recover from errors gracefully.

This proposal fixes the lag between constexpr exception support and library constexpr support and all future constexprification papers should make their exception types constexpr.

Implementation experience

Implemented in libc++ as part of implementation of P3068, with source code available on my github and with compiler + library available on the compiler explorer.

```
constexpr bool check(const char * msg) {
    try {
        auto vec = std::vector<int>{0,1,2,3
        return vec.at(4) == 4; // out-of-ra
    } catch (const std::out_of_range & exc)
```